Math 196 Problem Set 1

Due April 2, at 10:00 PM

For book problems, see the pdf on canvas or in my email.

Problem 1. Book Exercises 9 and 10 from Section 1.1.

Problem 2. Book Exercises 11 and 13 from Section 1.1.

Problem 3. Book Exercise 70 from Section 1.2.

Problem 4. Find a quadratic polynomial

$$f(t) = at^2 + bt + c$$

for some $a, b, c \in \mathbb{R}$, which satisfies $f(-1) = 1$ $f(1) = -1$, $f(2) = 4$.

Problem 5. Suppose we have a system of linear equations with more variables than equations.

(a) Show by example that it is possible for such a system to have no solutions.

(b) Argue that if such a system has at least one solution, it must have infinitely many.[1]

Problem 6. Consider the vectors:

$$v = \begin{bmatrix} -1 \\ 3 \end{bmatrix}, \; w = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Compute the following vectors, and plot them in the plane:

$$v - w, \; 3w - v, \; -2w$$

Problem 7. Consider the matrices:

$$A = \begin{bmatrix} 1 & 2 \\ 0 & -1 \\ 3 & 0 \end{bmatrix}, \; B = \begin{bmatrix} 1 & 0 & 2 & 4 \\ -1 & 1 & 1 & 1 \end{bmatrix}, \; C = \begin{bmatrix} 1 & 3 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & 1 \end{bmatrix}$$

For each matrix vector product below, determine whether it is well defined and if it is, compute it.

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \; A \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}, \; B \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \; B \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \; B \begin{bmatrix} 2 \\ 4 \\ 3 \\ -1 \end{bmatrix}, \; C \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

Problem 8 (Optional)**.** Find a matrix A so that for any x, y, z, we have

$$A \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} y \\ -z \\ x \end{bmatrix}$$

Problem 9. Consider the vectors

$$v = \begin{bmatrix} 2 \\ -1 \end{bmatrix}, \; w = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$$

In $\mathbb{R}^2$.

[1]Hint: Consider the reduced row echelon form of the coefficient matrix. How many pivot columns can there be?

(a) Using Gaussian elimination, find a way to express the vector

$$\begin{bmatrix} -3 \\ 1 \end{bmatrix}$$

as a linear combination of v and w.

(b) Argue, based on the reduced row echelon form you found in part (a), that in fact any vector in $\mathbb{R}^2$ can be written in a unique way as a linear combination of v and w. (Later, we will say that v and w form a *basis* of $\mathbb{R}^2$.)

(c) (Optional) Find an explicit formula in terms of x and y that expresses an arbitrary vector

$$\begin{bmatrix} x \\ y \end{bmatrix}$$

as a linear combination of v and w.

Problem 10. Let A be the matrix

$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}$$

Compute the matrix vector products

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \; A \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \; A \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \; A \begin{bmatrix} -1 \\ -1 \end{bmatrix}, A \begin{bmatrix} 1 \\ -1 \end{bmatrix},$$

Plot these points in the plane, and guess at a geometric description of transformation taking v to Av.